Since $\sqrt{x} \geq 0$, the selection of $\varepsilon < 1$ a fortiori (which of course, does not impair the generality of our proof) (even stronger reason) allows us to square the last inequalities $\boxed{(1-\varepsilon)^2 < x < (1+\varepsilon)^2}$

On removing the parentheses, we obtain $\boxed{(-2\varepsilon+\varepsilon^2) < (x-1) < (2\varepsilon+\varepsilon^2)}$ (4)

Note that inequalities (4) are equivalent to (3) (provided that $0 < \varepsilon < 1$). Now let us proceed from (4) to a more exacting inequality:

?? $\boxed{|x-1| < (2\varepsilon-\varepsilon^2) \quad (5)}$ (since $0 < \varepsilon < 1$, we have $(2\varepsilon-\varepsilon^2) > 0$).

It is easy to conclude that if (5) holds, inequalities (4) and, consequently, (3) will hold all the more. Thus, for an arbitrary ε within $0 < \varepsilon < 1$, it is sufficient to take $\boxed{\delta = 2\varepsilon - \varepsilon^2}$. ??

Reader: What happens if $\varepsilon \geq 1$?

Author: Then δ determined for any $\varepsilon < 1$ will be adequate a fortiori.

Reader: Apparently, we may state that $\lim_{x \to 2} \sqrt{x} = \sqrt{2}$, $\lim_{x \to 3} \sqrt{x} = \sqrt{3}$ and, in general, $\lim_{x \to a} \sqrt{x} = \sqrt{a}$.?

Author: Yes, that's right.

Reader: But could we generalize it to $\boxed{\lim_{x \to a} f(x) = f(a)}$.

Author: Yes, it is often the case. But not always. Because the function $f(x)$ may be undefined at point a. Remember that the limit of the function $x \sin \frac{1}{x}$ at point $x=0$, but the function itself is not defined at $x=0$.

Reader: But perhaps the equality $\lim_{x \to a} f(x) = f(a)$ can be considered as valid in all the cases when $f(x)$ is defined at point a?

Author: This may not be correct either. Consider, for example, a function which is called the "fractional part of x". The standard notation for this function is $\{x\}$. The function is defined on the whole real line. We shall divide the real line into half-intervals $[n, n+1[$. For x in $[n, n+1[$ we have